ELEVENLABS TRACK / MISSISSAUGA

Who gets heard by voice AI?

Measured speech gaps, where affected language communities live,
and a Cantonese callback prototype.

FLEURS • CENSUS • 311 • ELEVENLABS



Findings snapshot | November 2026

Exploratory analysis; not a resident-level failure estimate.

THE SIGNAL

A measurable gap—and a clear caveat

All 14 FLEURS language groups were processed; request-audio coverage is shown separately.

26.43%

Highest mean FLEURS CER

Cantonese • 40 clips

10,765

Cantonese home-language speakers

2021 census count

989 / 2,653

Clips transcribed

Cached Scribe results across the inventory

What we can say Cantonese has the highest CER in the completed FLEURS sample.

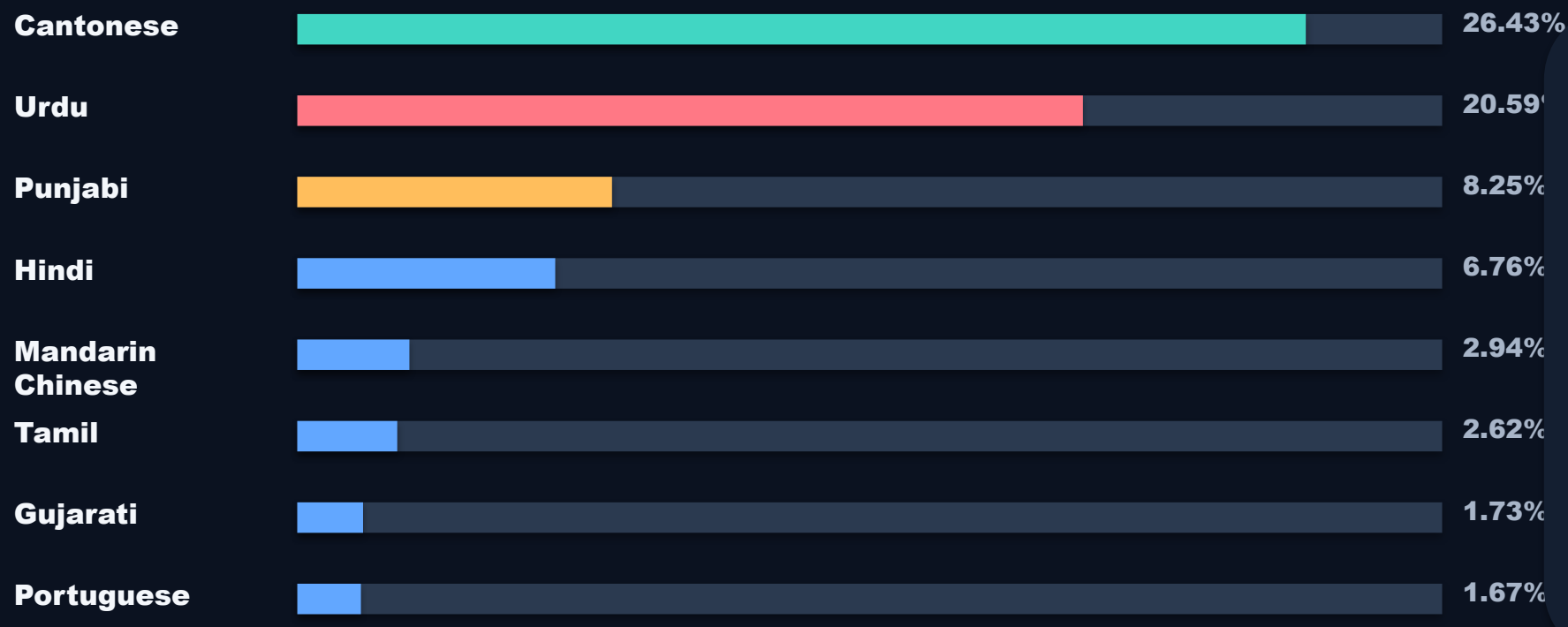
What we cannot say That 26% of Cantonese callers—or 10,765 residents—will be misunderstood.

Why FLEURS is read speech, with unmatched prompts and no speaker IDs; service-request audio is missing.

Sources: outputs/asr_summary.csv; 2021 census-derived language counts; outputs/REPORT.md.

Cantonese and Urdu stand out on CER

Mean character error rate across 40 FLEURS clips per language; lower is better.



Read this carefully

Mandarin and Cantonese WER is near 100% because word boundaries/tokenization differ.

CER is more informative here. Cantonese also has the highest CER, not just a WER artifact.

FLEURS is read speech —not live 311 calls.

Accent Archive: 419/468 English-accent clips processed.

Matched prompt ID 1589: one clip per language. It controls text, not speaker, gender, or dialect.

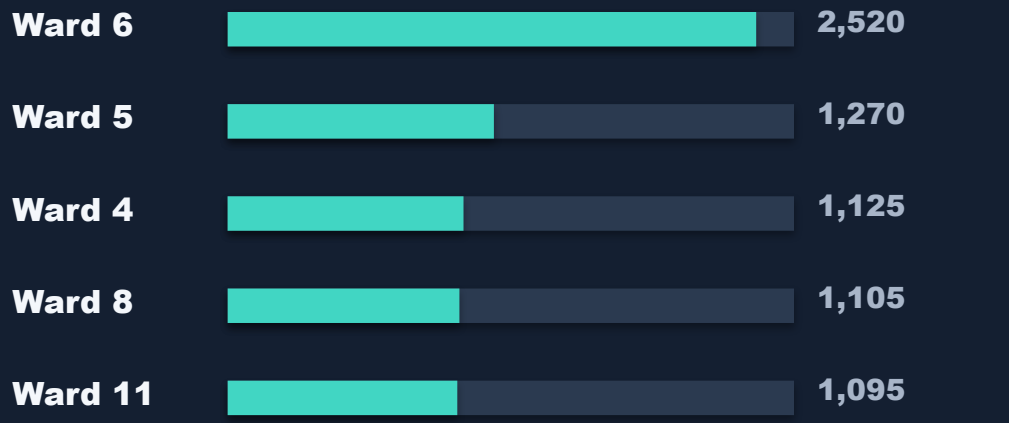
Source: outputs/asr_summary.csv and outputs/paired_fleurs_summary.csv; normalized CER/WER.

Cantonese speakers are present across the city

The census identifies potential reach for better language access—not individual outcomes.

Largest ward counts

10,765
citywide Cantonese speakers



Largest neighbourhood counts



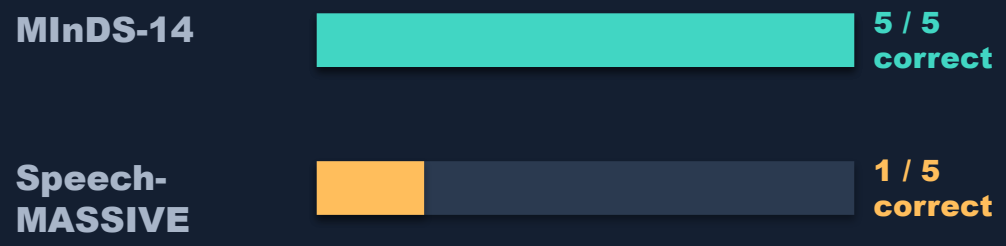
Counts are home-language speakers and may overlap with other language counts.

Source: 01_people_census/derived/geo_language_long.csv (corrected derived geography).

Transcription is only one part of understanding

The request-audio corpora are banking and voice-assistant data, not Mississauga 311 calls.

Audio → transcript → intent



Audio transcripts are classified against dataset labels. These remain banking and virtual-assistant tasks, not city services.

Separate text-only baseline

MInDS-14	90.6–98.1%
Speech-MASSIVE	55.2–62.1%

Classifier input is reference text—not Scribe audio output. Tasks are banking and virtual-assistant labels, not city services.

Design implication: confirm the interpreted request and offer a human/alternate-language fallback.

Sources: outputs/audio_intent_evaluation.csv; outputs/intent_evaluation.csv; sample sizes n=1 per audio language.

The index points to places for follow-up

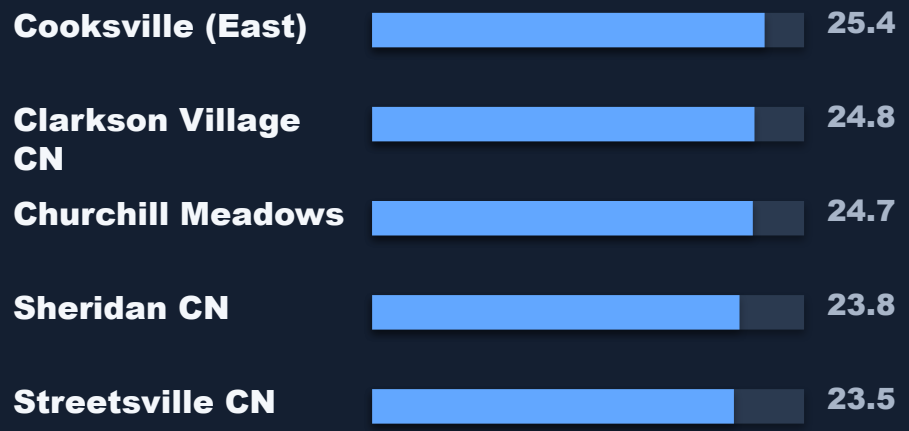
Higher scores combine language-group errors with census speaker counts; they are not failure probabilities.

Wards

Index score



Populated neighbourhoods

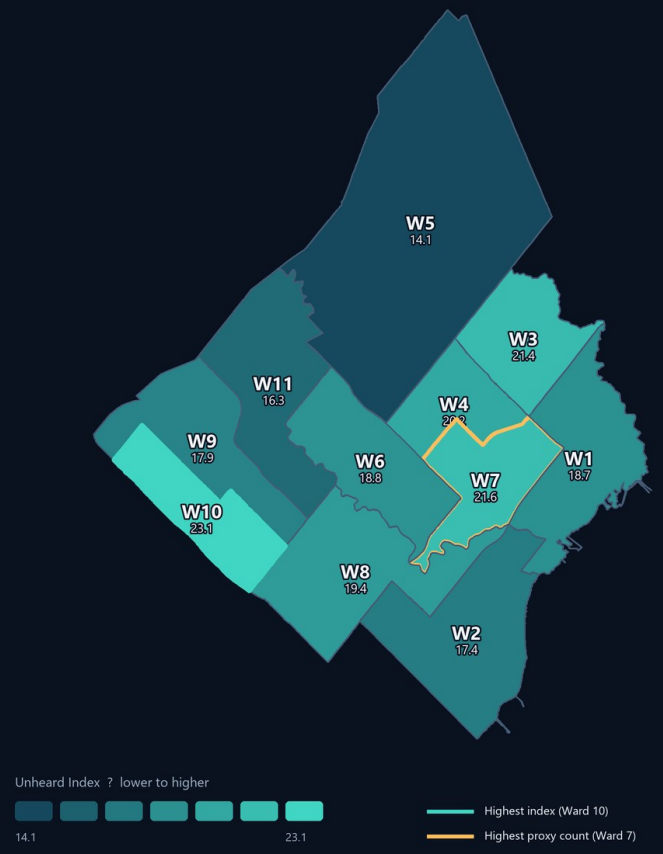


Ward 7 has the largest ward-level weighted-error proxy (4,133.6); index leaders differ from proxy-count leaders.

Source: outputs/unheard_index.csv; 50 geographies. Full FLEURS CER + limited audio-intent evidence.

A geographic view of the exploratory index

Ward boundaries with population-weighted benchmark signals; not observed caller failures.



23.096

Highest ward index

Ward 10 • 23.31% coverage

4,133.6

Largest proxy count

Ward 7 • not known failed calls

Neighbourhood signal

Cooksville (East) • index 25.449 • 11.96% coverage

Interpret with care

The index combines available FLEURS CER and pilot intent errors. The request-audio pilot has only one clip per language; scores are not service-level outcomes or resident-level risk.

Boundaries: City of Mississauga Ward Boundaries service. Scores: 2021 census-derived outputs/unheard_index.csv; verify boundary crosswalk.

A Cantonese callback proof of concept

TTS gives the measured gap a tangible response—but it does not prove service effectiveness.

CANTONESE / TRADITIONAL CHARACTERS

你好，我哋收到你嘅請求。

“Hello, we received your request.”

Audio: outputs/311_callback_yue.mp3 • 39,749 bytes

Before any real use

1 Fluent Cantonese review of translation and pronunciation.

2 Test comprehension with community participants.

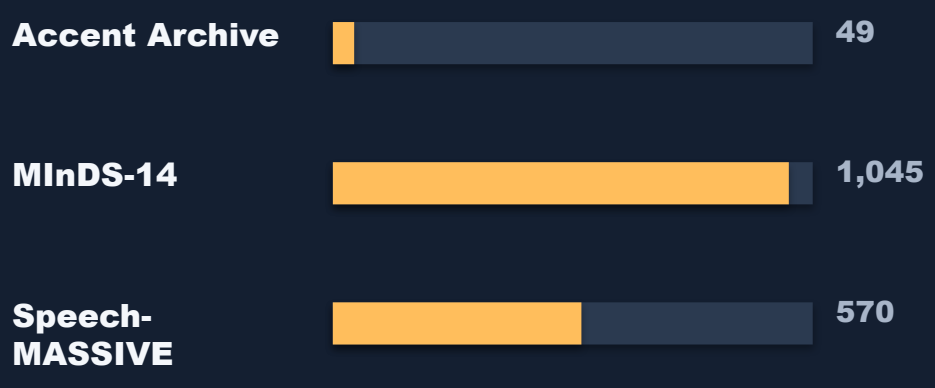
3 Keep human assistance available; TTS is not automatic translation.

NEXT STEPS

The evidence is useful—and still incomplete

The next phase should improve coverage before making service-level claims.

Remaining Scribe inventory



1,664 clips remain

Cached results are preserved for resume.

Recommended next

01 Expand consented 311-style speech examples in missing languages.

02 Validate transcripts and routing with fluent speakers.

03 Test translation, callback comprehension, and human fallback.

04 Revisit geography scores with service-specific outcome evidence.

Treat these findings as a prioritization signal—not as a count of residents already misunderstood.